

SPINRISE ERP

Web Form Blueprint

Enterprise UI / UX Standards for ERP Migration

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Company	Kalpatharu Software Ltd
Classification	Confidential — Internal Use Only

1. Design Vision & Objective

SpinRise ERP is the structured migration of Kalpatharu Software Ltd's legacy VB6-based Kalsofte ERP to a modern, React-based web application. The migration is not a cosmetic re-skin — it is a systematic re-engineering of every transaction screen to eliminate the usability pain points accumulated over 20+ years in ERP 7.4, while retaining the business logic and field vocabulary the operations team already knows.

1.1 Benchmark

The correct benchmark for every SpinRise screen is: Does this screen solve the pain points of an ERP 7.4 user in a spinning mill with fewer keystrokes, less training time, and fewer errors? Oracle Fusion and Zoho Books are NOT the benchmark.

1.2 Core Design Principles

Principle	Description
Tier-1 B2B Standard	Every screen must meet the visual and usability standard of leading enterprise ERP products. 'Acceptable' is not the goal — 'Wow factor' is.
Modern Web-Native Design	SpinRise must NOT copy the VB6 visual style. Retain domain logic and field vocabulary users know; deliver a clean, modern, web-native visual standard.
Single-Screen Architecture	Every transaction form must fit within one screen with zero vertical or horizontal scrolling. CEO-level closed decision.
Keystroke Efficiency	The number of keystrokes per transaction must be minimised, counted, and reported as an efficiency metric. The software must be operable without IST support.
WYSIWYG Fidelity	The screen display must accurately represent the transaction printout at all times.
Context-Sensitive Inputs	On each keystroke, the system loads only matching values. Related fields appear adjacent and auto-populate in sequence.
Audit Readiness	Every form must carry complete audit control — who, what, when — visible at a glance.

2. Global UI Theme Standards

2.1 Color Palette

All hex values below are authoritative for SpinRise ERP. Do not use colours outside this palette without a formal change request.

Token	HEX	Usage	Accessibility Note
Primary Blue	#185FA5	Primary buttons, links, active menu, section headers	WCAG AA on white (contrast 5.6:1)
Dark Blue	#0C447C	Headings, cover page, hover state on primary	WCAG AAA on white
Blue 100	#E6F1FB	Selected row highlight, active input border tint	Background use only
Background	#F5F5F3	App/page background	Non-interactive surface
Surface Alt	#FAFAF8	Section header bands, alternate table rows	Non-interactive surface
Border	#E2E2E2	Field borders, table separators, dividers	Structural only
Text 900	#1A1A1A	Primary body text, labels, data	WCAG AAA on white
Text 600	#4A4A4A	Secondary labels, helper text	WCAG AA on white
Text 400	#888888	Placeholder text, metadata	WCAG AA on white (large text only)
Success	#3B6D11	Approved status, success notifications, positive values	WCAG AA
Success Light	#EAF3DE	Success badge background	Background use only
Warning	#BA7517	Warning alerts, attention states	WCAG AA
Warning Light	#FAEEDA	Warning badge background, note boxes	Background use only
Error	#A32D2D	Validation errors, rejected status	WCAG AA
Error Light	#FCEBEB	Error badge background, error field tint	Background use only
Purple	#722ED1	PO Raised status badge	WCAG AA

2.2 Typography Standards

Primary font: Inter (web) / Segoe UI (system fallback). Monospace: JetBrains Mono (code, reference numbers).

Usage	Size	Weight	Line Height	Example
Page Title / H1	22 px	Extra Bold 800	1.2	SpinRise ERP Web Form Blueprint
Section Heading / H2	16 px	Bold 700	1.3	2. Global UI Theme Standards
Sub-heading / H3	14 px	Semi Bold 600	1.35	2.1 Color Palette
Body Text	13 px	Regular 400	1.5	All hex values below are authoritative...
Field Label	12 px	Semi Bold 600	1.35	Purchase Order Number
Caption / Metadata	11 px	Regular 400	1.5	Last updated: 08 May 2026
Table Header	11 px	Semi Bold 600	1.35	Token HEX Usage
Table Data	12 px	Regular 400	1.5	#185FA5
Monospace (ref no.)	11 px	Medium 500	1.4	PR-2026-00329
Badge / Status	11 px	Semi Bold 600	1.0	Approved

2.3 Spacing & Layout Standards

All spacing is derived from a 4 px base unit. ERP screens must use compact spacing — excessive whitespace wastes screen real estate and forces scrolling.

Token	Value	Application
sp-1	4 px	Icon padding, tight badge padding
sp-2	8 px	Field internal padding (top/bottom), inline gap
sp-3	12 px	Label-to-field gap, small section gap
sp-4	16 px	Field horizontal padding, card padding
sp-6	24 px	Section body padding, major content gap
sp-8	32 px	Between form sections, card gap
Field Height (compact)	32 px	Standard ERP input, date picker, select
Field Height (small)	28 px	Grid cell inputs, compact toolbar controls
Row Height (grid)	32 px	Transaction line item grid
Top Navigation Height	48 px	Fixed application header bar
Sidebar Width	220 px (expanded) / 60 px (collapsed)	Module navigation
Page Max Width	1280 px	Maximum layout width on wide screens

3. Application Layout Blueprint

3.1 Top Navigation Bar

The top bar is fixed (sticky) at 48 px height. It must display the company name as the primary identity element — not a search bar.

Element	Position	Specification
Company Name	Left	Kalpatharu Software Ltd — 13 px, Semi Bold, #1A1A1A
SpinRise Logo	Left	Next to company name, 24 px height, CEO-approved mark only
Division Name	Left	Shown below company name — 11 px, #4A4A4A
Branch Selector	Centre	Dropdown: current branch — 12 px, compact (28 px height)
Financial Year	Centre	Display current financial year — 12 px
Notification Bell	Right	Icon only; badge count on unread notifications
User Profile	Right	Avatar + name chip; dropdown with settings / logout
Theme Toggle	Right	Light / Dark switch — compact icon button
Logout	Right	Icon button with tooltip

Sticky behavior: the top bar must remain visible at all scroll positions. Background: #FFFFFF with 1 px border-bottom #E2E2E2 and shadow-sm.

3.2 Sidebar Navigation

Compact collapsible sidebar. Default: expanded (220 px). Collapsed: icon-only (60 px). Module groupings follow the SpinRise module matrix.

Module Group	Modules	Icon
Purchase	Purchase Requisition, Purchase Order, Arrival Entry	ShoppingCart
Inventory	Raw Material, Finished Goods, Transfers	Package
Sales	Sales Order, Delivery, Returns	TrendingUp
Finance	Accounts Payable, Receivable, Journal	DollarSign
HR	Attendance, Payroll, Shift Schedule	Users
Admin	User Management, Roles, Settings	Settings

Active menu: background #185FA5, text white, left border 3 px #0C447C. Hover: background #E6F1FB. Inactive text: #4A4A4A. Icon size: 18 px.

4. Dashboard Blueprint

The dashboard is the first screen after login. It must deliver high-density, actionable information in a compact layout. Every element must be justified by operational value.

4.1 KPI Strip (CEO Confirmed — 5 Cards)

CEO Decision (06 May 2026): Exactly 5 compact KPI cards in a single horizontal strip. All 5 must be visible without scrolling. Cards: Total Lines | Total Quantity (UOM grouped) | Approx. Budget | Created By | Approval Status.

Add Mode Suppression Rule — See Section 5.1.1 below for mode-aware display requirements.

Card	Value Type	Format	Icon
Total Lines	Count	Integer	List
Total Quantity	Multi-UOM	Nos: 241 Mtr: 42 Kg: 18	Package
Approx. Budget	Currency	₹ 2,45,800.00 (2dp)	DollarSign
Created By	Text	Employee name + date	User
Approval Status	Status	Status badge (see Section 11)	CheckCircle

Card height: max 72 px. Card background: white. Border: 1 px #E2E2E2. Border-radius: 8 px. Shadow: shadow-sm. No animation on load — static render.

4.2 Dashboard Panels

Panel	Description	Priority
Approval Queue	Documents pending user's action — sorted by urgency	High
Activity Timeline	Recent transactions by the logged-in user (last 10)	Medium
Quick Actions	New PR / New PO / View List — large tap-friendly buttons	High
Pending PRs	PRs awaiting conversion to PO — count + list	Medium
Notification Feed	System alerts, approvals received, rejections	Medium

5. Web Form Design Standards

This is the most critical section of the blueprint. Every SpinRise transaction form must conform to all rules below. No exception is permissible without a formal amendment approved by the CEO.

5.1 Single-Screen Architecture

CEO-level closed decision: Every SpinRise transaction form — header, line item grid, approval status, KPI strip, and footer — must be fully visible and operable on one screen with zero vertical or horizontal scrolling at 1366 × 768 minimum resolution.

Design Element	Rule
KPI Strip	Compact single-line horizontal strip — max 72 px height
Form Header	Max 2 rows of fields — use multi-column layout (3–4 cols)
Line Item Grid	Virtualized — shows 8–12 rows visible; scrolls within its own container only
Approval Status	Compact read-only band below header — not a full-width multi-node block
Action Buttons	Sticky bottom bar — Save, Submit, Print, Cancel — always visible

5.1.1 Add Mode Display Rule

CEO Decision (06 May 2026 — Finding #29): KPI cards must be mode-aware.

In Add mode: KPI strip shows only Total Lines and Approx. Budget. Created By, Approval Status, and Approval Pipeline are hidden. After first save: All three fields activate with real data and remain visible in Edit/View modes.

Rationale: These three fields have no meaningful value before initial document save. Hiding them in Add mode declutters the form and improves keystroke efficiency for new document creation.

Implementation: Frontend renders KPI strip conditionally:

- Add mode: Show cards [Total Lines, Approx. Budget] only
- Edit/View mode: Show all 5 cards

5.2 Label Standards

CEO Correction (06 May 2026): Labels must be Title Case — NOT ALL CAPS. All-caps reduces readability on web forms. No abbreviations permitted.

Rule	Correct	Incorrect
Title Case — no abbreviations	Purchase Order Number	PO NO / PO no. / PO NUMBER
Title Case — no abbreviations	Approved Quantity	Appr. Qty / QTY
Title Case — no abbreviations	Required Date	Req. Date / REQ DATE

Title Case — no abbreviations	Approximate Cost	Appr. Cost / APPROX COST
Title Case — no abbreviations	Department Name	Dept. / DEPARTMENT
Full descriptive label	Unit of Measure	UOM / U.O.M
Mandatory indicator	* Purchase Order Number	Purchase Order Number (no *)
Label size / weight	12 px, Semi Bold 600, #1A1A1A	14 px Bold / 10 px Regular
Label position	Above the field (top-aligned)	Inside field as placeholder only
Hyphens in labels	Not permitted	Purchase-Order-Number

5.3 Field Alignment Rules

Data Type	Alignment	Applies To
Numeric (quantity, count)	Right	Quantity, Total Lines, Record Count
Numeric (currency / value)	Right	Rate, Unit Price, Value, Approx. Cost, Total Value
Text / varchar	Left	Item Code, Description, Vendor Name, Department
Date	Left	PR Date, Required Date, Delivery Date
Dropdown / select	Left	Status, UOM, Department, Section
Status badges	Centre	Approval Status column in grids

This standard applies to every field in every form across all SpinRise modules. No exceptions.

5.4 Decimal Precision Standards

Kalsoft ERP Standard — applies to all SpinRise modules. Quantity: 3 decimal places | Rate: 4 decimal places | Value: 2 decimal places.

Field Type	Decimal Places	Example Display	Formula
Quantity	3	241.500	User input
Rate (Unit Price)	4	₹ 125.7500	[DEFERRED] Last PO Rate / Avg Rate / Manual Entry (with Justification field). To be confirmed in Blueprint v1.1 after FSD amendment approved by CEO.
Value	2	₹ 30,368.25	Quantity × Rate (editable when rate manually entered, otherwise auto-calculated)
Percentage	2	12.50 %	User input
Cost (Landed)	2	₹ 30,750.00	Value + Tax + Insurance + Freight + Other charges

Value field width must be wide enough to display large numeric values without truncation (minimum 120 px).
Unit Price / Rate field must not display the currency symbol twice — fix immediately.

5.4.1 Multi-Vertical Field Configuration Standard

REFERENCE IMPLEMENTATION: Budget Category Field

- Field Name: Budget Category
- Data Type: Dropdown
- Visibility: PR Type = Capital AND Division Profile = Paper Mill
- Mandatory when visible

Architecture Pattern:

This conditional display logic establishes the reference pattern for all future vertical-specific fields in SpinRise. All such fields must follow this structure:

- Visibility rule explicitly documented in Blueprint
- Backend configuration table maps field to division + transaction type
- Frontend renders conditionally based on logged-in user's division profile

NOTE: Sasi to confirm backend configuration table design before this section is finalised and Blueprint is submitted for CEO sign-off.

5.5 Tab Sequence & Mandatory Field Enforcement

Tab sequence validation is enforced at the UI layer — not just server-side.

Rule	Behavior
Mandatory field guard	If a mandatory field is empty, the cursor must not advance to the next tab stop. UI must display inline error immediately.
Auto-advance on Tab	Tab from last field of a grid row auto-creates the next row. Cursor lands on the first input of the new row.
Grid initialisation	When user clicks 'New Document', the first item row loads automatically. Cursor lands on Item Code of row 1.
Add Line button	Used only for adding subsequent rows after the auto-created first row.
Skip navigation	Tab must skip read-only and display-only fields. Only editable fields participate in tab sequence.
Form section order	Header fields → Line item grid → Approval status (read-only) → Action buttons.
Add Multiple button	Opens a combined interaction with two modes — both must be available: Mode 1 — Multi-Select Item Lookup (browse users): Opens item search modal. User selects multiple items using checkboxes. Each selected item loads as a separate individual line row in the grid. Mode 2 — Quick Row-Count Input (experienced/power users): Prompts for a number (N). Inserts N blank rows. Cursor lands on Item Code field of the first new row. User tabs through to fill each row. Named Customer Requirement: Pallvaa Mills, SKS

(Requirement originated by Sasi in FSD review — Finding #15 and #16, R2.0)

5.6 Input Control Standards

Text Fields

Property	Value
Height	32 px (compact ERP standard)
Border	1 px solid #E2E2E2; border-radius: 6 px
Focus state	2 px solid #185FA5; background #F0F7FF
Placeholder	Short hint text only — not a substitute for a label
Error state	1 px solid #A32D2D; background #FCEBEB; inline message below
Disabled state	Background #F5F5F3; text #888888; cursor: not-allowed

Dropdowns & Select Controls

Property	Value
Width	Dynamic — fits content; minimum 160 px
Search	Searchable on first keystroke — no need to open then type
Loading	Async loading with debounce (300 ms) — shows spinner
Virtual scroll	Required for lists > 100 items
Max visible	8 items before scroll within dropdown

Switch Controls (preferred over Checkbox)

Use Switch instead of Checkbox for all boolean fields. Switch provides clearer active/inactive visual state for ERP power users. Checkbox is permitted only in multi-select grids.

Date Controls

Property	Value
Keyboard input	Must support direct typing (DD/MM/YYYY or DD-MM-YYYY)
Calendar picker	Compact — opens on field click or keyboard trigger
Default value	PR Date auto-populates with current login date (editable by user)
Financial date lock	Option to restrict dates within current financial year
Display format	DD-MMM-YYYY (e.g., 08-May-2026) — human readable

5.7 Context-Sensitive Filters

Filters must be context-sensitive — not static dropdowns of all values.

Pattern	Behavior
Keystroke-based loading	On each keystroke in a lookup field, API call returns only matching values. Debounce: 300 ms.
Parent-child chaining	Department field populates → Section field filters to show only sections within selected department. Fields appear adjacent.
Item lookup	Inline search icon inside Item Code field. Keyboard: Alt+Down or Ctrl+Space opens lookup modal. F1 is browser-reserved — do NOT override.
Filter priority	Filter decisions for each form must be documented and confirmed before next demo. (Action: Palanivel — filter decision document by Sat 09 May.)

5.8 Item Image — Item Master Integration

Item thumbnail must appear in exactly two locations:

Location	Behavior	Size
PR line item grid	Displayed in a fixed column when item is entered or selected	28 × 28 px thumbnail
Item lookup modal	Thumbnail column in search results; hover shows larger preview	36 × 36 px (grid) / 120 × 120 px (hover)

Both images are fetched from Item Master. Mariyaiya to add Item Image as a formal field in FSD amendment.

5.9 PR List Screen — Companion Document Register

CEO Decision (07 May 2026 — Finding #18): The PR List Screen is a new FSD screen requirement accepted by CEO as part of the PR module scope.

COMPANION SCREEN: PR List Screen

A document register screen is required for the Purchase Requisition module. This screen displays all PR documents with the following confirmed features:

- Approval Status badge column (colour-coded per Section 10 badge standards)
- Filter by status
- Pagination

Pending items (not yet included in this Blueprint version):

- Full column list — Seenivasan and Palanivel's input due Fri 8 May 2026 to Mariyaiya. Formal column specification to be added in Blueprint v1.2 after FSD amendment is approved and IST column requirements confirmed.

Timeline: List screen specification will be formally included in Blueprint v1.2, post Sasi review and CEO final approval of this version.

6. Lookup Modal Design Standard

The item lookup modal is the primary navigation point for all master data selection in SpinRise. It replaces the F1-key behaviour from ERP 7.4.

SpinRise Lookup Standard (CEO Closed Decision): Inline search icon inside Item Code field. Keyboard alternative: Alt+Down or Ctrl+Space opens the lookup modal. F1 is reserved by the browser (Chrome / Edge / Firefox) for native Help — cannot be overridden. Do NOT report F1 as a defect in any SpinRise screen during IST testing.

Modal Element	Specification
Trigger	Inline search icon in field Alt+Down Ctrl+Space
Modal size	720 × 480 px; centered overlay; dark semi-transparent backdrop
Search bar	Auto-focused on open; debounced (300 ms); searches code, description, alias
Filter row	Category, UOM, Active/All — compact filter chips above the grid
Result grid columns	Item Code Description UOM Rate Image (thumbnail) Stock
Row height	36 px with thumbnail; 32 px without
Pagination	Virtual scroll preferred; fallback 50 items per page
Multi-select	Checkbox column for multi-select; Add Selected button loads selected items as individual rows. Quick row-count input inserts N blank rows. Pattern for Add Multiple button in Section 5.5.
Quick preview	Hover on row shows enlarged item image + full spec in floating tooltip
Keyboard nav	Arrow Up/Down to navigate; Enter to select; Escape to close
Performance	Results must appear within 300 ms for datasets up to 10,000 items

7. Enterprise Table Grid Standards

The transaction line item grid is the most operationally critical UI element in SpinRise. It must handle large datasets with inline editing, keyboard navigation, and row-level validation. The grid component choice has significant long-term impact on developer productivity and user experience.

7.1 Grid Component Comparison

STATUS: PROPOSED — Pending CEO Decision (Fri 8 May 2026)

Note: The comparison table and grid design standards in Section 7.3 that do not depend on library selection (row height, column alignment, field types, etc.) are CONFIRMED. The library selection itself (AG Grid Enterprise vs alternatives) is PENDING CEO approval.

Feature	AG Grid Enterprise	Ant Design ProTable	TanStack Table v8	MUI DataGrid Pro
Virtual Scrolling	Excellent	Good	Good (manual)	Good
Inline Cell Editing	Native — rich	Partial	Manual build	Good
Keyboard Navigation	Excel-like	Partial	Manual build	Partial
Frozen Columns	Yes	Yes	Yes	Yes
Row / Column Grouping	Yes (built-in)	Limited	Manual	Limited
Column-Level Filtering	Rich (multi)	Good	Manual	Good
Bulk / Range Editing	Yes	No	Manual	No
Excel-Style Copy-Paste	Yes	No	No	No
Export (Excel / CSV)	Built-in	Via EPPlus	Manual	Pro tier
Row-Level Validation	Yes	Partial	Manual	Partial
10,000+ Row Performance	Excellent	Good	Excellent	Good
TypeScript Support	Excellent	Good	Excellent	Good
Bundle Size	~420 KB	~180 KB (AntD)	~15 KB	~200 KB
Licence	Enterprise (paid)	Free (MIT)	Free (MIT)	Paid (Pro)
ERP Readiness Rating	★★★★★	★★★☆☆	★★★☆☆	★★★★☆

7.2 Recommendation — AG Grid Enterprise

Proposed: AG Grid Enterprise — pending CEO final decision.

AG Grid Enterprise is the only option that provides Excel-like copy-paste, bulk range editing, and true column grouping out of the box. For a spinning mill ERP where power users enter 40–60 line items per transaction with keyboard-only workflows, these are non-negotiable capabilities. The licence cost is justified by the elimination of months of custom grid development work.

Why Ant Design ProTable is insufficient for large ERP grids:

- No bulk/range editing — users must click each cell individually
- No Excel-style copy-paste — major productivity loss for high-volume data entry
- Keyboard navigation is partial — Tab across cells is not native
- Row-level validation requires custom implementation

- Suitable for simple list views and small grids (< 200 rows); not transaction entry grids

7.3 Grid Design Standards (Applied to All Modules)

Property	Standard
Row height	32 px compact (ERP standard)
Header height	36 px; sticky — always visible regardless of scroll within grid
Frozen columns	Item Code, Item Description — always visible on horizontal scroll
Inline editing	Click or Tab into cell to edit; validation on blur
Required Date column	Minimum 90 px width (CEO confirmed)
Number columns	Right-aligned; 3 dp (Qty) / 4 dp (Rate) / 2 dp (Value)
Text columns	Left-aligned; truncate with tooltip on overflow
Status column	Centred; status badge (see Section 11)
Image column	28 × 28 px thumbnail; centred
Selection column	Checkbox; 40 px wide; frozen left
Row highlight on hover	Background #E6F1FB
Alternate row shading	Background #F0F5FF on even rows
Error row highlight	Background #FCEBEB; red left border 3 px

7.4 Grid Footer — Summary Row

Footer Element	Calculation	Format
Total Lines	COUNT(rows)	Integer — right-aligned
Total Quantity	SUM grouped by UOM	Nos: 241 Mtr: 42 Kg: 18
Total Value	SUM(Quantity × Rate) per line	₹ 2,45,800.25 (2 dp)
Total Approx. Cost	SUM(Rate + Tax) per line	₹ 2,75,000.00 (2 dp)

Footer is sticky — remains at the bottom of the grid container at all times. Background: #F5F5F3. Border-top: 2 px solid #185FA5.

8. Validation & Error Standards

Validation Type	Trigger	Display	Color
Required field empty	On blur / Tab attempt	Inline below field: 'This field is required'	Error red #A32D2D
Invalid format	On blur	Inline below field: 'Enter a valid date (DD-MM-YYYY)'	Error red #A32D2D
Duplicate document	On Save / Submit	Alert banner at top of form	Error red #A32D2D
Quantity zero	On blur	Inline: 'Quantity must be greater than 0'	Error red #A32D2D
Over-budget warning	On value calculation	Inline warning below value field	Warning amber #BA7517
Success save	On Save complete	Toast notification top-right, auto-dismiss 3 s	Success green #3B6D11
Mandatory field guard	On Tab / Enter	Field border turns red; cursor stays; no advance	Error red #A32D2D

Error Summary Panel: When form submission fails, a summary panel appears at the top listing all errors with field names as anchor links. This prevents users from hunting for errors across a large form.

9. Notifications & Alerts

Notification Type	Trigger	Auto-Dismiss	Placement
Success	Save / Submit / Approve completed	3 seconds	Top-right toast
Information	System message, process started	5 seconds	Top-right toast
Warning	Near-limit, potential issue	8 seconds	Top-right toast
Error	Save failed, validation error	No — manual	Top-right toast + inline
Confirmation	Delete / Submit / Reject action	N/A	Centred modal dialog
Approval received	Another user approved your document	Manual	Notification bell + banner

Do not use browser `alert()` / `confirm()` / `prompt()` — replace with `Ant Design Modal.confirm()` and notification API.

10. Status Badge Standards

Status badges must be consistent across all SpinRise modules. Colors below are CEO-confirmed.

Status	Background	Text Color	Usage
Requested	#E6F1FB	#185FA5	PR / PO raised, awaiting first approval
Pending	#FAEEDA	#BA7517	Waiting for an intermediate approver
Approved	#EAF3DE	#3B6D11	Fully approved, cleared for next step
Rejected	#FCEBEB	#A32D2D	Rejected at any stage — requires revision
PO Raised	#F0E6FB	#722ED1	PR has been converted to a Purchase Order
Cancelled	#F5F5F3	#4A4A4A	Document cancelled by authorised user
Partial	#FAEEDA	#BA7517	Partially approved / partially delivered
Closed	#E8E8E8	#4A4A4A	Transaction fully completed and closed

Badge style: font-size 11 px, font-weight 600, border-radius 4 px (pill shape: 9999 px), padding 2 px 8 px.
Minimum width 72 px for visual consistency across columns.

11. Print & Export Blueprint

PDF generation: QuestPDF (CEO confirmed). Excel/CSV export: EPPlus (CEO confirmed). No other reporting libraries in Phase 1.

11.1 PDF Print Layout (A4)

Print Region	Specification
Page size	A4 portrait; margins: 15 mm top, 15 mm bottom, 18 mm left, 15 mm right
Print header	Company logo Company Name Document Title Document Number PR Date PR Time (adjacent to date)
Print footer	Total Value Date + Time stamp Page X of Y Form name (NOT in red)
Line item table	Same columns as screen grid; quantity 3 dp; rate 4 dp; value 2 dp
Decimal format	Kalsoft standard — Qty 3dp, Rate 4dp, Value 2dp — enforced on all print output
Total row	Printed at bottom of last page; includes Total Quantity (UOM grouped) + Total Value
Approval section	Printed below items — shows approval chain with names, designations, dates, signatures
Watermark	DRAFT on unsubmitted documents; APPROVED on approved documents
Item image	Printed beside item description if image exists in Item Master
Font	Arial or Helvetica (QuestPDF compatible); 9 pt body; 7 pt captions

11.2 WYSIWYG Fidelity

WYSIWYG (What You See Is What You Get): The print output must match the screen layout at all times. Column order, decimal precision, label text, and field values on the screen must match the printed document exactly. No discrepancies between screen and print are acceptable.

12. Accessibility & Keyboard Standards

Standard	Specification
Contrast ratio	Minimum WCAG 2.1 AA — 4.5:1 for body text; 3:1 for large text and UI components
Keyboard navigation	All interactive elements reachable via Tab; all actions triggerable via keyboard
Focus indicator	2 px solid #185FA5 outline; border-radius matching control; never hidden (no outline: none)
F1 key	RESERVED by browser (Chrome / Edge / Firefox) for native Help. Cannot be overridden. Do NOT use F1 for item lookup in SpinRise.
Lookup shortcut	Alt+Down or Ctrl+Space opens item lookup modal — documented as SpinRise standard across all modules
Tab sequence	Follows visual reading order left-to-right, top-to-bottom; skip read-only fields
Error announcement	Validation errors must be programmatically associated with fields (aria-describedby)
Screen readability	All interactive elements must have accessible names (aria-label or visible label)

Colour-only information	Status must never be conveyed by colour alone — always pair with text label or icon
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13. Recommended Technology Stack

The table below shows our current stack, what we retain, and what we propose to add — with a clear justification for each change. No change is proposed without an ERP-specific reason.

13.1 Full Stack Comparison

Layer	Current Stack	Proposed	Action	Rationale
UI Framework	Ant Design 5	Ant Design 5	RETAIN	Current stack — under review for grid component selection. Not explicitly CEO-approved independent of component library decision.
Runtime	React 18	React 18	RETAIN	Stable; concurrent features not yet needed
Language	TypeScript	TypeScript (strict)	RETAIN	Type safety critical for ERP data integrity
Build Tool	Vite	Vite	RETAIN	Fastest HMR; no reason to change
Routing	React Router v7	React Router v7	RETAIN	Stable; file-based routes work well
State (global)	Zustand	Zustand	RETAIN	Simpler than Redux Toolkit for team size; sufficient for current complexity
HTTP Client	Axios	Axios	RETAIN	Interceptor pattern works well for auth token injection
Form State	Manual useState	React Hook Form	ADD	ERP forms have 30-50+ fields. RHF uses uncontrolled inputs — 60-70% fewer re-renders. See Section 13.2.
Validation	Ad-hoc logic	Zod + RHF resolver	ADD	Type-safe schema validation; errors at compile time not runtime. See Section 13.2.
Data Grid	Ant Design Table	AG Grid Enterprise	PROPOSED	Full keyboard nav, inline editing, Excel copy-paste for large grids. Pending CEO decision. See Section 7.
PDF Print	QuestPDF	QuestPDF	RETAIN	CEO-confirmed for all PDF output
Excel Export	EPPlus	EPPlus	RETAIN	CEO-confirmed for all Excel/CSV export
Icons	Ant Design Icons	Lucide Icons	ADD	Lighter weight, tree-shakeable, consistent stroke style
Backend	ASP.NET Core 8	ASP.NET Core 8	RETAIN	Clean architecture; Dapper + SQL Server SPs
Database	SQL Server 2016	SQL Server 2019	LEVERAGE	Blueprint must document SQL 2019 features available: JSON_VALUE, STRING_AGG, TRIM, improved indexing, APPROX_COUNT_DISTINCT

13.2 Why Add React Hook Form + Zod

Short summary for CEO review:

Problem (Current)	Solution (Proposed)	Benefit
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Manual useState for every field causes re-renders on each keystroke. A 40-field PR form re-renders 40+ times per keystroke.	React Hook Form uses uncontrolled inputs — fields are registered, not controlled. No re-render until submit or explicit trigger.	60–70% reduction in renders. Noticeably smoother UI on slow machines common in factory environments.
Ad-hoc validation is scattered across event handlers. Decimal rules, mandatory checks, and business rules live in different places.	Zod schema defines all validation rules in one place. zodResolver connects it to React Hook Form automatically.	Single source of truth for validation. Runtime errors caught at compile time. Consistent error messages across all forms.
No standard for form submission state — loading spinners and error states are manually managed per form.	React Hook Form provides formState.isSubmitting, errors, dirtyFields — all computed automatically.	Reduces boilerplate by ~30% per form. Developers write less code, make fewer mistakes.

Migration effort: Low. React Hook Form and Zod are added alongside the existing stack — no refactor of existing forms required before they can be used on new forms.

13.3 SQL Server 2019 — Features to Leverage

Feature	Use Case in SpinRise
STRING_AGG	Concatenate PR line items or approver names in single-row summaries without cursor loops
JSON_VALUE / JSON_QUERY	Return structured data from SPs for dynamic configuration (module settings, user preferences)
TRIM()	Replace LTRIM(RTRIM()) — cleaner SP code for string sanitisation
APPROX_COUNT_DISTINCT	Fast cardinality estimates for large dashboards (total customers, total items)
Improved statistics	Automatic index maintenance improvements — better query plans for large transaction tables
In-Memory OLTP	For high-frequency audit log writes — optional Phase 2 optimisation

14. UX Best Practices for ERP Power Users

SpinRise must be operable without IST support. The following practices must be applied to all screens to achieve that standard.

Practice	Implementation
Minimise keystrokes	Auto-populate PR Date (login date), Sample (No), first grid row. Count keystrokes per transaction and report as efficiency metric.
Smart defaults	Default values must be contextually meaningful — Sample=No, Date=Today, Currency=INR, UOM from Item Master. PR Type default is customer/division-configurable at implementation time (not hardcoded).
Context-sensitive filters	Dropdowns load only relevant values based on prior selections (Department → Section → Cost Centre).
No popup overload	Maximum one modal open at a time. Confirmations only for destructive or irreversible actions.
Preserve user context	Navigation must not reset form state. Use route state or session storage for in-progress documents.
Inline help	Field-level tooltips for non-obvious inputs. No need for a separate help page for standard fields.
Auto-save drafts	Draft saved automatically every 60 seconds for transaction forms. Warn user on close if unsaved.
Undo / Redo support	Where technically feasible — especially for grid cell editing to prevent accidental data loss.
Approval cycle display	Compact read-only approval pipeline band showing current step — not a full-width multi-node diagram.
Rate selection clarity	FSD gap: Last PO Rate / Average Rate / Manual Entry (with Justification field). Decision pending Mariyaiya FSD amendment.
PR Type Default	Customer / division-configurable at implementation time. Not hardcoded in the application. Set per customer profile during implementation setup. Default value varies by customer and division.

15. UI/UX Screen Approval Cycle

No SpinRise screen demo will be accepted for CEO review unless it has passed through the full UI/UX review cycle in the sequence below.

Stage	Reviewer	Responsibility	Deadline Rule
Stage 1	Developer (Abinandan)	Self-check against this blueprint. All blueprint rules must pass before submission.	Before submission to IST
Stage 2	IST Member (Muthuvel)	Functional and usability review. Raise findings in Palanivel's scored format.	Within 2 working days
Stage 3	TL-IST (Palanivel)	Consolidated IST review. Use scored, prioritised, finding-numbered format with effort estimates.	Within 1 working day
Stage 4	PM (Saravanan)	Business completeness and FSD compliance check.	Within 1 working day
Stage 5	CEO (T. Mani)	Final design approval. Once cleared, screen is the official SpinRise UI standard.	As scheduled

Palanivel's review format (scored, prioritised, finding-numbered, effort-estimated, sprint-bucketed) is the template for all SpinRise UX review reports. Vignesh to align MOM format accordingly.

16. CEO Executive Summary

SpinRise ERP is not a VB6 form running in a browser. It is a professionally engineered, modern web application that addresses two decades of accumulated ERP 7.4 pain points — with fewer keystrokes, less training time, fewer errors, and a visual standard that earns customer confidence on first impression.

Dimension	Current State	Target Standard
Visual Quality	Functional — meets requirements	Tier-1 B2B with 'Wow factor' — Kalpatharu flagship product
Single Screen	Not yet achieved — some scroll present	Zero scroll at 1366×768 — CEO closed decision
Keystroke Count	Not measured	Counted, reported, and continuously reduced per transaction
Decimal Standard	Inconsistent across fields	3/4/2 dp uniform across all modules — Kalsoft standard
Print Quality	Basic output — some formatting issues	WYSIWYG — screen matches print exactly every time
IST Independence	IST support required for many workflows	Zero IST support needed for daily operations
Label Standard	Mixed — some CAPS, some abbreviations	Title Case, full descriptive, no abbreviations
Grid Capability	Basic Ant Design Table	AG Grid Enterprise (proposed) — keyboard nav + inline editing
Review Cycle	Ad-hoc demo submissions	Formal 5-stage review: Dev → IST → TL-IST → PM → CEO

With this blueprint as the governing standard, SpinRise ERP will deliver a consistent,

auditable, high-performance user experience across all 14 modules — setting the foundation for Kalpatharu Software Ltd's next 20 years of ERP delivery.

Document End — SPINRISE ERP Web Form Blueprint v1.1

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